NMC 2013 O-LEVEL PAPER 1 QUESTIONS

SECTION A : 5 marks each

Qn. 1. If $xy = 2, yz = 24, zx = 3$ and x, y and z are all positive numbers, what is the value of $x+y+z$?

Qn. 2. In the sum below, the letter $F = 0$ and the other letters represent the digits $1, 2, 3, 4, 5$ or 6 with each digit used exactly once. The two-digit integer AB is a prime number. What is the value of $A + B$?

$$\begin{array}{r} AB \\ +CD \\ \hline EFG \end{array}$$

Qn. 3. If a and b are non zero numbers such that a and b are the two roots of $x^2 + ax + b = 0$. What is the value of a and b?

Qn. 4. Evaluate $\sqrt[3]{\frac{1\times2\times4+2\times4\times8+3\times6\times12+4\times8\times16+\cdots}{1\times3\times9+2\times6\times18+3\times9\times27+4\times12\times36+\cdots}}$

Qn. 5. Suppose that $f(x + 1, y) = f(x, y) + y + 1$, $f(x, 0) = x$ and $f(x, y) = f(y, x)$ for all x and y. What is the value of $f(12, 4)$?

Qn. 6. Find all values of k such that the equation $\log(kx) = 2\log(x + 1)$ has exactly one root.

Qn. 7. Four pupils are standing in front of a closed room and they want to open it. Each makes a statement. One statement is false, and the other three are true.
Ann: "I do not have the key and Cal does not have the key."
Ben: "I do not have the key and Ann does not have the key."
Cal: "I do not have the key and Ben does not have the key."
Don: "I do not have the key and Ann does not have the key."
Who actually has the key?

Qn. 8. What is the last digit y of the 9-digit number $19700019y$ given that the number is prime?

Qn. 9. For any real number x, let $[x]$ be the greatest integer less or equal to x for example if $x = 4.9$ then $[x] = 4$. Let m and n be positive real numbers such that $m \cdot [m] = 17$ and $n \cdot [n] = 11$. What is the value of m and n?

Qn.10. A boy has as many sisters as he has brothers. Each of his sisters has twice as many brothers as she has sisters. How many children are there in the family?

SECTION B : 10 **marks each**

Qn.11. In our cafeteria, each cake costs an integer number of coins and also each sandwich costs an another integer number of coins. If every girl in the class buys one cake and every boy buys one sandwich, they will spend one coin less than if every boy buys one cake and every girl buys one sandwich. There are more girls than boys in the class. How many more girls are there?

Qn.12. Lydia has got five real numbers a, b, c, d, e where $a < b < c < d < e$. She takes the sum of each pair of numbers and writes down the ten sums. The three smallest sums are $32, 36$ and 37, while the two largest sums are 48 and 51. Determine e.

Qn.13. (a) A bricklayer would take 9 hours to build a certain wall on his own and another would take 10 hours to build the same wall. If the two work together, they sometimes stop for a chat so that they lay a total of 10 fewer bricks per hour. If it takes them 5 hours to build the same wall together, what is the total number of bricks in the wall?

(b) During a recent period of time, 11 days had some rain. A morning rain was always followed by a clear afternoon and an afternoon rain was always preceded by a clear morning. A total of nine mornings and twelve afternoons were clear. How many days had no rain at all?

Qn.14. Two positive numbers x and y are given. If the sum $x + y$, the product xy and the difference $x^2 - y^2$ are all equal. Find the value of y.

Qn.15. Let $ABCD$ be a rectangle and let P be a point inside the rectangle. If $\overline{PA} = 8, \overline{PB} = 4$ and $\overline{PD} = 7$. What is the value of $\overline{PC}$?

NMC 2013 O-LEVEL PAPER 2 SOLUTIONS

SECTION A : 5 marks each

1. The difference between the two plants' height is 36 cm currently. The smaller plant grows 9cm in 6 years, and the taller plant grows 5cm in 6 years. Thus, the smaller plant can make up 4cm in each 6 years. Hence, it will take $(\frac{36}{9} = 9) \times 6$ years, that is, 54 years for the two plants to be equally high.

2. Among the given distances, the largest one equals the sum of all the others: $8 = 2+1.5+3.5+1$. This is possible only if the sheds are along a line. The order of the sheds must be C, B, A, E, D. Therefore, distant(B, E) = distant(B, A) + distant$(A, E) = 2 + 1 = 3$km.

3. The sum of the numbers on all the faces is 21. The first observation implies that the sum of the numbers on a couple of opposite sides is 8. This couple can be $(6, 2)$ or $(5, 3)$. The second observation implies that the sum of the numbers on another couple of opposite sides is 9. That couple can be $(6, 3)$ or $(5, 4)$. Combining the above information, the only possibility is that $(6, 2)$ and $(5, 4)$ are on opposite sides. Hence, $(3, 1)$ is the third couple of opposite sides, therefore the answer is 3.

4. The mathematician Augustus lived his entire life during the 1800's. In the last year of his life he stated: Once I was x years old in the year x^2. In which year was Augustus born?.

5. **section B**. Find the total number of three-digit numbers that are divisible by 9 and contain no even digits ?
   A number is divisible by 9 if and only if the sum of its digits is divisible by 9, so we need to find three odd digits whose sum is 9 or 27. (The sum of three odd numbers is odd, so 18 is impossible.) The possibilities for the digits are:

   (i) $1, 1, 7$ which can be arranged to give three different numbers which are $117, 171$ and 711.

   (ii) $1, 3, 5$ which can be arranged to give six different numbers. These are $135, 153, 315, 351, 513$ and 531.

   (iii) $3, 3, 3$ which gives only one number,333.

   (iv) $9, 9, 9$ which also gives one number 999. The total is therefore $3 + 6 + 1 + 1 = 11$.

6. One can of orange paint contains $\frac{3}{5}$ of a can of red paint and $\frac{2}{5}$ of a can of yellow paint. One can of green paint contains $\frac{2}{3}$ of a can of blue paint and $\frac{1}{3}$ of a can of yellow paint. When orange and green are mixed, we have $(\frac{2}{5} + \frac{1}{3})$ of a can of yellow paint in a total of 2 cans, so the proportion is $(\frac{2}{5} + \frac{1}{3})/2 = \frac{11}{30}$.

7. A restaurant can accommodate at most 400 people, which includes the guests and the waiters. Each waiter can serve a maximum of 12 guests. What is the greatest number of guests that can be served ?
   solution

We need to know how many sets of 13 people (12 guests and 1 waiter) can be formed from 400 people. Now $400 = 30 \times 13 + 10$. But those last 10 people in the remainder also need a waiter. Therefore 31 waiters are required, which means $400 - 31 = 369$ guests can be served.

8. Nine numbers are written in ascending order. The middle number is also the average of the nine numbers. The average of the five largest numbers is 68 and the average of the five smallest numbers is 44. What is the value of the middle number?
Solution
Let the sum of all the numbers be S, so the middle number is $\frac{1}{9}S$. The sum of the five largest numbers is $5 \times 68 = 340$, and the sum of the five smallest numbers is $5 \times 44 = 220$. If we add these sums together, then we have counted the middle number twice, so $S + \frac{1}{9}S = 340 + 220$, that is $\frac{10}{9}S = 560$. Thus $S = \frac{560 \times 9}{10} = 504$. Therefore the middle number is $\frac{1}{9}S = \frac{1}{9} \times 504 = 56$.

9. Tammy, John,and Martha were all born at noon on January 19th, but in different years. When Tammy was four years old,John was three times as old as Martha. When Martha was twice as old as Tammy, John was five times as old as Tammy. How old was Tammy when John was twice as old as Martha?
Solution sec B Let j denote Johns age when Tammy was 4, and let m denote Marthas age when Tammy was 4. Let a be the number of years after Tammy was 4 when Martha was twice as old as Tammy and John was five times as old as Tammy. Let b be the number of years after Tammy was 4 when John was twice as old as Martha. Then we have the following equations: $(i)j = 3m, (ii)m + a = 2(4 + a), (iii)j + a = 5(4 + a), and (iv)j + b = 2(m + b)$. We want to know the value of $4 + b$. From (iv),$b = j - 2m$ so that, from (i), $b = m$. From (ii), $m = 8 + a$. Using (i) in (iii), we also have $3m = 20 + 4a$.So $b = m = 4m3m = 4(8 + a)(20 + 4a) = 12$. Hence,$4 + b = 16$.

10. Each of the following five statements is either true or false. How many of these statements are true given that statement (4) is true.

    (1) Statements (3) and (4) are both true.

    (2) Statements (4) and (5) are not both false.

    (3) Statement (1) is true.

    (4) Statement (3) is false.

    (5) Statements (1) and (3) are both false.

    solution

    If Statement (4) is true, then Statement (3) is false. Hence, (3) and (4) cannot both be true. It follows that Statement (1) cannot be true. Hence, both Statement (1) and Statement (3) must be false. We see then that Statement (4) and Statement (5) are true. It follows that Statement (2) is true. The answer is 3. Thus three statements are true.

11. The two shortest sides of a right triangle have lengths $\sqrt{3}$ and 2. Let α be the smallest interior angle of this triangle. What is the value of sin α?
Solution

The side opposite to angle α has length $\sqrt{3}$ and the hypotenuse of the triangle is $\sqrt{(\sqrt{3})^2 + 2^2} = \sqrt{7}$. Therefore $\sin \alpha = \frac{\sqrt{3}}{\sqrt{7}} = \sqrt{\frac{3}{7}}$.